

DELHI PUBLIC SCHOOL, BOKARO STEEL CITY

Unit Test — Biology

Class: X

Time: 1 hour

Maximum Marks: 48

Answer all questions. Marks for each question are indicated against it. Draw diagrams neatly in pencil.

1. Which blood vessel carries blood away from the heart? [2]
2. Which of the following organelles is primarily involved in photosynthesis? [2]
3. Explain the role of chloroplasts in photosynthesis, naming the main pigments involved and briefly outlining the two major stages of the process. [2]
4. Describe the flow of blood through the human heart starting from the right atrium and ending at the aorta; include the names of valves crossed. [2]
5. Draw a labelled diagram of an alveolus showing capillaries and air space (label alveolar sac, capillary, and direction of gas exchange). [2]
6. Draw a neat labelled diagram of the human digestive system (stomach, small intestine, large intestine, liver, pancreas) and label the site where most absorption occurs. [5]

7. Draw and label a nephron (Bowman's capsule, glomerulus, proximal tubule, loop of Henle, distal tubule, collecting duct). [5]
8. Explain the structural differences between palisade mesophyll and spongy mesophyll and state how each structure aids its function in the leaf. [5]
9. Describe the process of transpiration in plants in two to three sentences and name two environmental factors that increase its rate. [5]
10. Explain how the structure of xylem vessels facilitates water transport in plants (mention one structural feature and its role). [5]
11. A diagram shows two potted plants — Plant A in bright light with broad green leaves, Plant B kept in dim light with pale, elongated leaves.
 - (a) Account for the difference in leaf colour between Plant A and Plant B. [2]
 - (b) Suggest one practical measure to help Plant B recover. [3]
12. A resting person has tidal volume (air per breath) of 0.5 L and breathes 12 times per minute. Calculate the pulmonary ventilation rate. [5]
13. If dead space is 0.15 L per breath, calculate the alveolar ventilation per minute. Show working. [5]